

YANGDOK, Korea, Democratic People's Republic of

WMO: 470520

Lat: 39.217N

Lon: 126.650E

Elev: 915

StdP: 14.22

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.6	0.2	-9.7	3.4	0.8	-5.9	4.2	3.6	18.7	15.1	17.3	16.2	0.6	320	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	87.7	74.9	85.3	73.7	83.0	72.3	78.0	84.5	76.6	82.3	75.3	80.2	5.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.2	141.4	81.9	74.9	135.2	80.1	73.7	129.8	78.7	42.3	84.7	40.9	82.6	39.6	80.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.5	15.3	13.2	-9.1	91.9	5.8	2.5	-13.2	93.7	-16.6	95.2	-19.9	96.5	-24.1	98.3		
			-9.8	80.9	5.2	2.0	-13.6	82.4	-16.6	83.6	-19.5	84.7	-23.3	86.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	18.2	23.4	35.4	47.9	59.6	68.3	74.2	74.4	64.6	51.9	36.8	22.5		
	DBStd	20.81	8.42	8.17	7.14	6.97	5.41	4.07	4.50	4.43	5.04	6.60	8.79	9.05		
	HDD50	3620	987	744	456	119	4	0	0	0	0	57	400	853		
	HDD65	6882	1452	1164	919	514	178	15	2	2	67	406	845	1318		
	CDD50	2971	0	0	2	56	302	548	750	755	437	116	5	0		
	CDD65	759	0	0	0	0	11	113	287	293	54	1	0	0		
	CDH74	5357	0	0	0	20	304	912	1904	1899	309	9	0	0		
	CDH80	1296	0	0	0	2	41	209	510	511	23	0	0	0		
	WSAvg	3.9	3.4	4.1	4.8	4.9	4.8	3.9	3.5	3.2	3.2	3.4	4.0	3.7		
	PrecAvg	43.6	0.5	0.7	1.0	2.2	3.5	4.8	12.5	10.3	4.4	2.0	1.7	0.9		
Precipitation	PrecMax	72.0	1.3	2.6	4.5	4.6	7.9	11.6	40.7	35.1	12.4	6.4	4.7	2.6		
	PrecMin	23.3	0.0	0.0	0.1	0.2	0.9	1.2	2.1	1.0	0.4	0.2	0.2	0.1		
	PrecStd	11.7	0.4	0.6	1.0	1.2	1.9	2.7	6.9	7.5	3.2	1.5	1.1	0.6		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.2	49.5	63.7	77.1	84.0	88.5	91.6	90.9	82.8	74.9	62.6	46.4		
		MCWB	36.0	43.0	49.4	57.6	61.9	68.1	76.6	77.6	70.1	60.6	53.6	42.3		
	2%	DB	37.3	43.3	57.9	71.6	80.1	85.2	87.7	87.6	79.9	70.6	57.5	42.2		
		MCWB	33.1	36.5	46.7	54.2	61.3	67.7	75.7	77.2	67.1	59.3	50.7	38.4		
	5%	DB	34.8	39.5	53.0	67.6	77.4	82.2	85.2	85.2	77.6	67.3	53.9	38.7		
		MCWB	31.2	34.2	43.3	53.0	59.7	66.7	74.9	75.8	65.7	56.9	48.5	35.7		
	10%	DB	32.5	36.4	48.5	63.4	74.0	79.3	82.6	82.8	75.3	64.1	50.6	35.6		
		MCWB	29.6	32.1	40.3	50.1	58.5	65.9	74.2	74.4	64.7	55.2	45.9	32.8		
	0.4%	WB	36.7	45.0	52.8	61.5	67.5	72.9	79.9	80.3	73.6	63.8	55.3	43.4		
		MCDB	40.1	48.0	60.5	72.8	77.6	82.3	87.5	86.9	78.6	71.0	59.8	45.8		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.9	37.7	48.0	56.8	63.8	70.9	78.0	78.4	70.2	60.6	51.9	38.9		
		MCDB	36.6	42.0	56.3	67.2	75.0	79.8	84.4	85.2	75.4	68.0	56.4	41.5		
	5%	WB	31.8	34.7	44.0	54.0	62.0	69.3	76.7	77.1	68.4	58.7	49.2	35.7		
		MCDB	34.5	38.6	51.5	64.1	72.7	77.7	82.2	83.4	74.6	64.9	53.0	38.4		
	10%	WB	29.6	32.6	41.0	51.3	60.2	68.1	75.5	75.7	66.6	56.8	46.6	33.2		
		MCDB	32.4	36.1	47.7	61.0	70.1	75.8	80.4	81.0	72.7	62.8	50.1	35.5		
Mean Daily Temperature Range	5% DB	MDBR	20.5	20.3	19.3	23.2	23.4	19.3	13.3	14.2	19.8	22.2	17.5	17.9		
		MCDBR	20.0	22.7	27.6	31.3	31.3	26.7	18.7	18.3	23.0	26.5	22.4	17.7		
		MCWBR	16.1	16.4	16.9	16.0	13.6	11.3	8.5	9.0	11.1	14.8	15.5	14.5		
		MCDBR	17.7	19.7	25.5	26.3	25.6	19.6	14.0	15.5	17.5	21.9	19.2	16.8		
	5% WB	MCWBR	15.2	14.8	16.5	14.9	13.0	9.6	7.4	8.9	10.0	13.5	14.5	14.2		
Clear-Sky Solar Irradiance	taub		0.392	0.463	0.531	0.602	0.614	0.685	0.648	0.554	0.485	0.501	0.460	0.398		
	taud		1.965	1.836	1.743	1.653	1.669	1.590	1.723	1.939	2.045	1.888	1.916	1.994		
	Ebn at Noon		230	229	228	222	223	207	214	232	240	215	204	217		
	Edh at Noon		42	55	66	77	77	84	73	57	49	51	44	38		
All-Sky Solar Radiation	RadAvg		788	1046	1354	1551	1738	1616	1282	1346	1317	1080	725	658		
	RadStd		49	71	124	133	167	154	180	160	133	75	84	52		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	+106

Nomenclature: See separate page